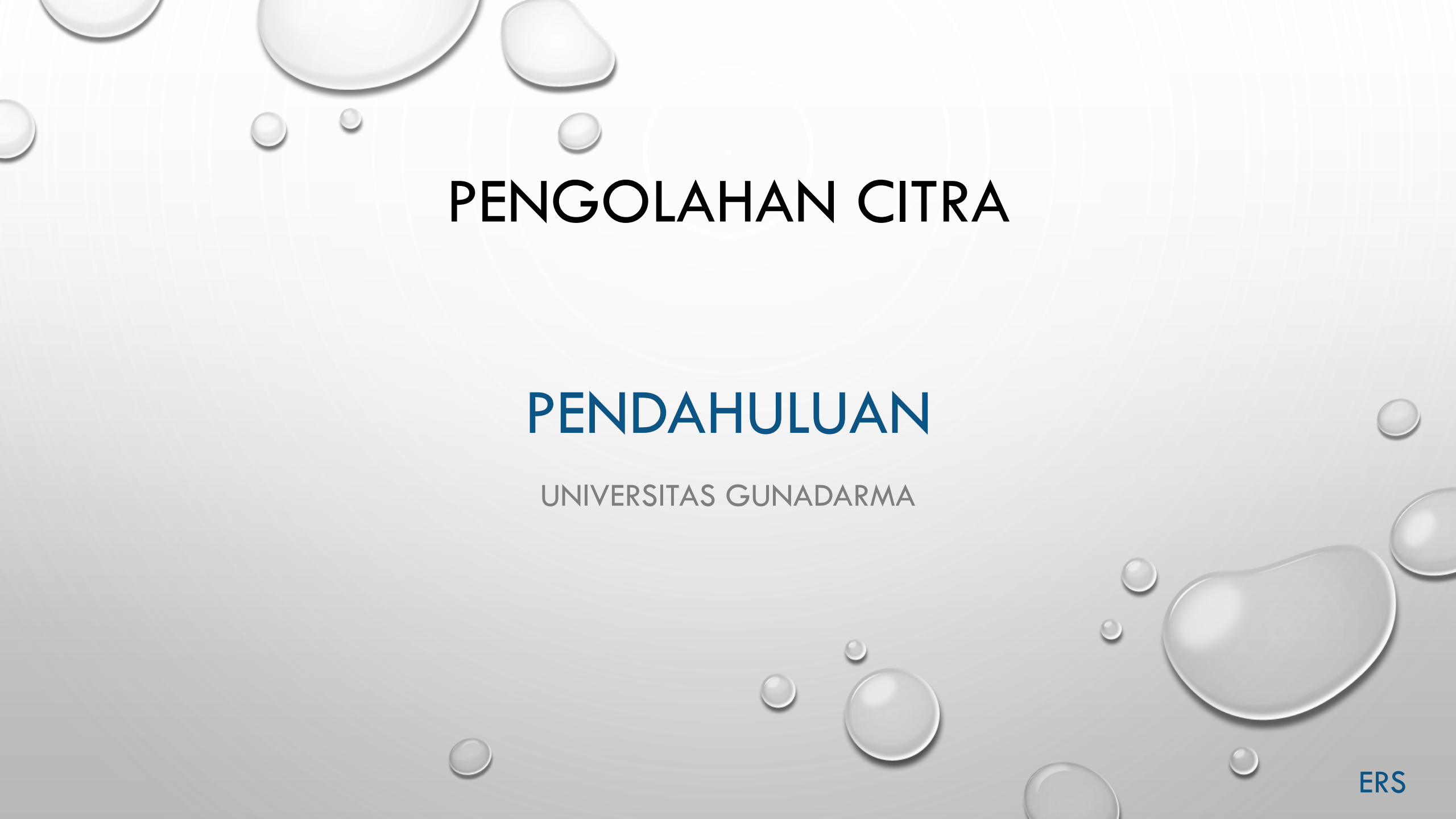




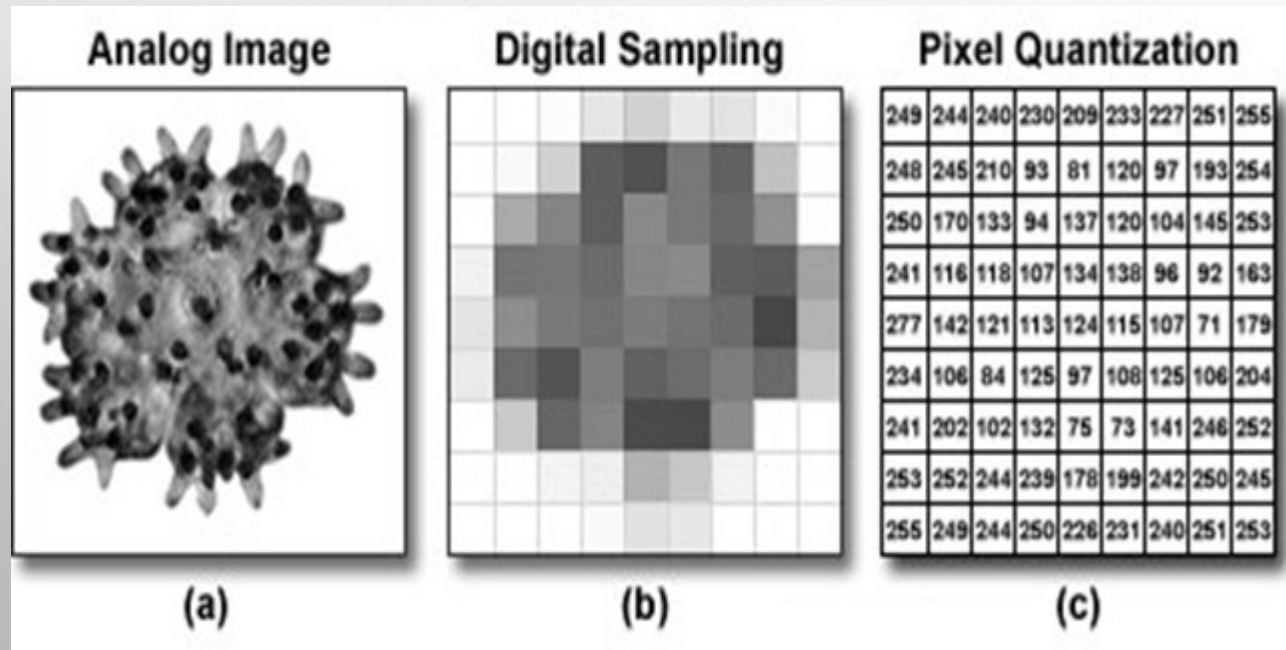
PENGOLAHAN CITRA

PENDAHULUAN

UNIVERSITAS GUNADARMA

DEFINISI CITRA DIGITAL

A DIGITAL IMAGE IS A REPRESENTATION OF A REAL IMAGE AS A SET OF NUMBERS THAT CAN BE STORED AND HANDLED BY A DIGITAL COMPUTER. IN ORDER TO TRANSLATE THE IMAGE INTO NUMBERS, IT IS DIVIDED INTO SMALL AREAS CALLED PIXELS (PICTURE ELEMENTS). FOR EACH PIXEL, THE IMAGING DEVICE RECORDS A NUMBER, OR A SMALL SET OF NUMBERS, THAT DESCRIBE SOME PROPERTY OF THIS PIXEL, SUCH AS ITS BRIGHTNESS (THE INTENSITY OF THE LIGHT) OR ITS COLOR. THE NUMBERS ARE ARRANGED IN AN ARRAY OF ROWS AND COLUMNS THAT CORRESPOND TO THE VERTICAL AND HORIZONTAL POSITIONS OF THE PIXELS IN THE IMAGE.



$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & f(1,2) & \dots & f(1,N-1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f(M-1,0) & f(M-1,1) & f(M-1,2) & \dots & f(M-1,N-1) \end{bmatrix}$$

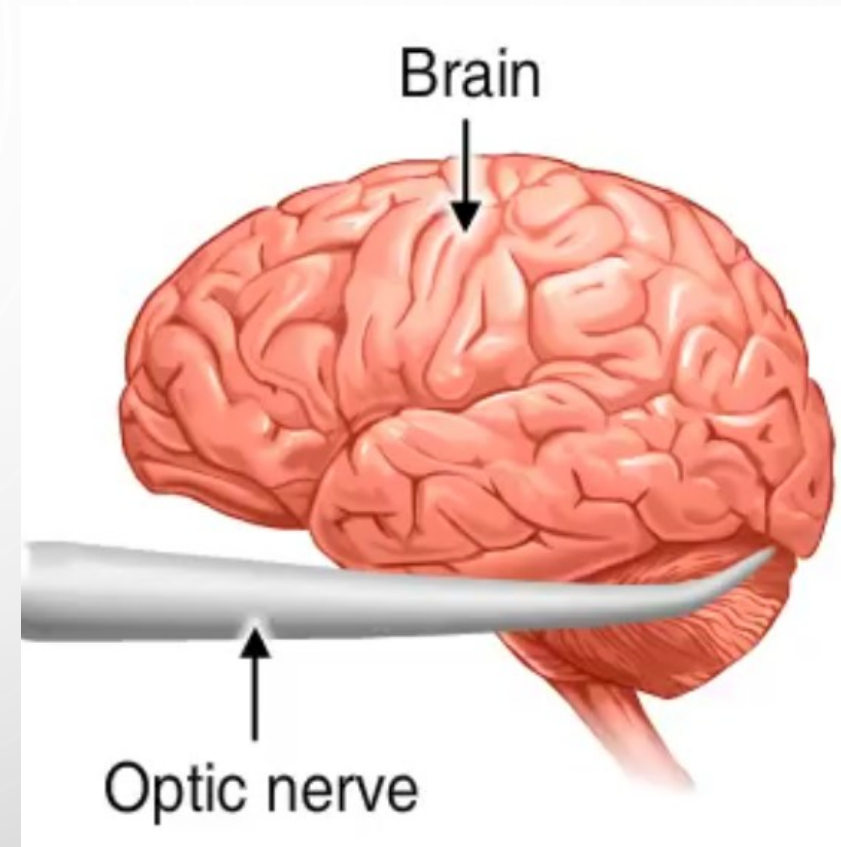
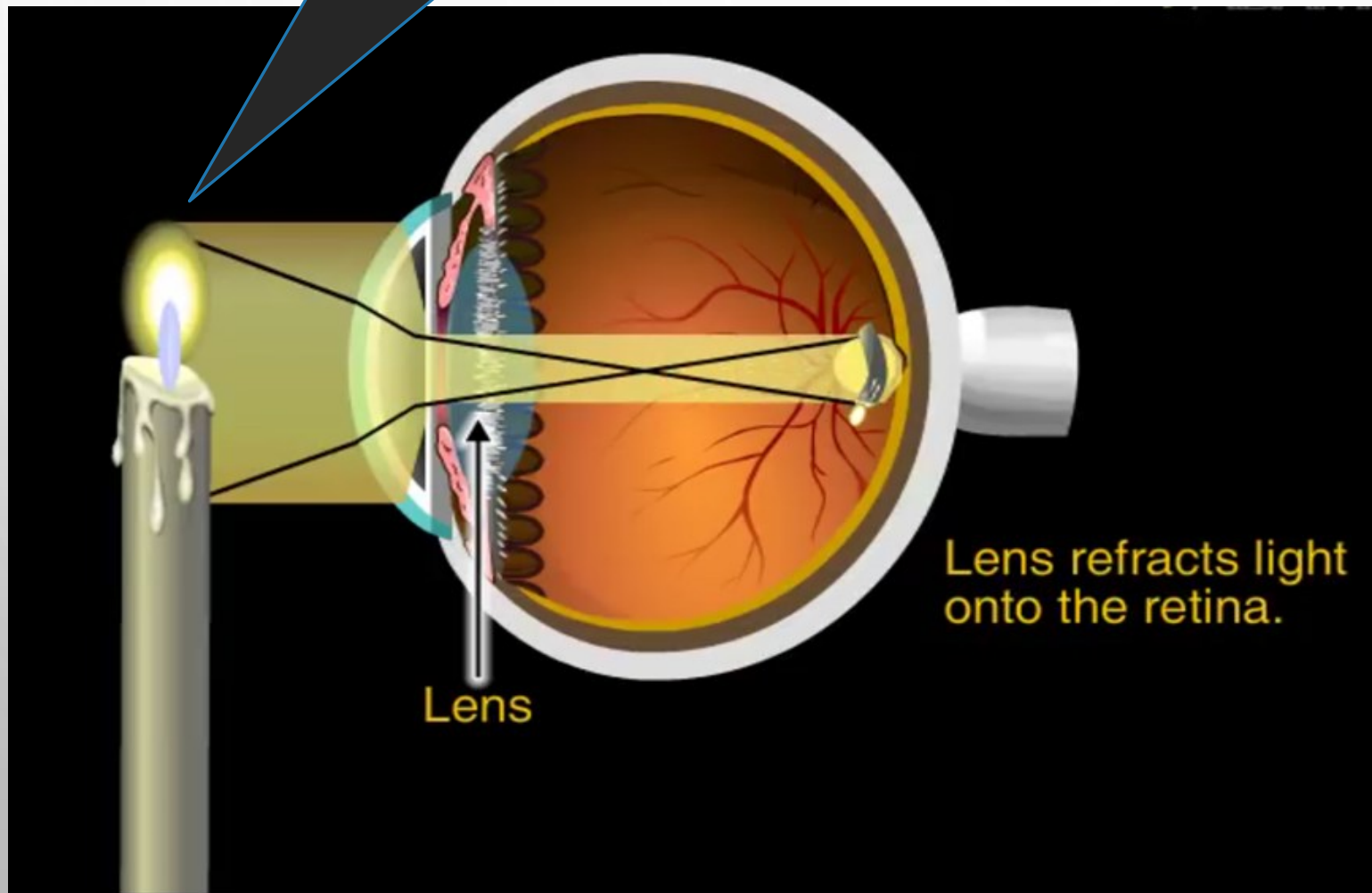
HOW DO WE STORE DATA INTO COMPUTER?

- 🕒 BINARY
- 🕒 Numeric?
- 🕒 Text??
- 🕒 Image???

KENAPA KITA BISA **MELIHAT** OBJEK ?

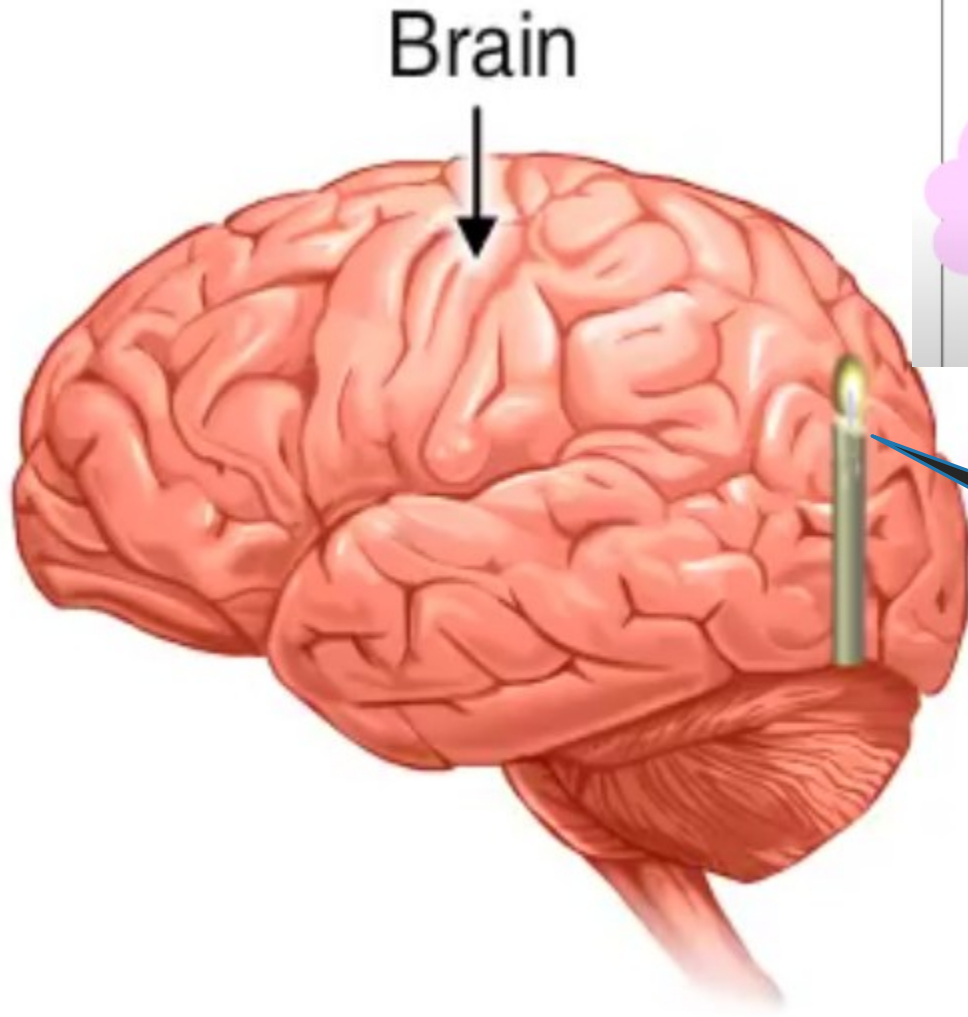
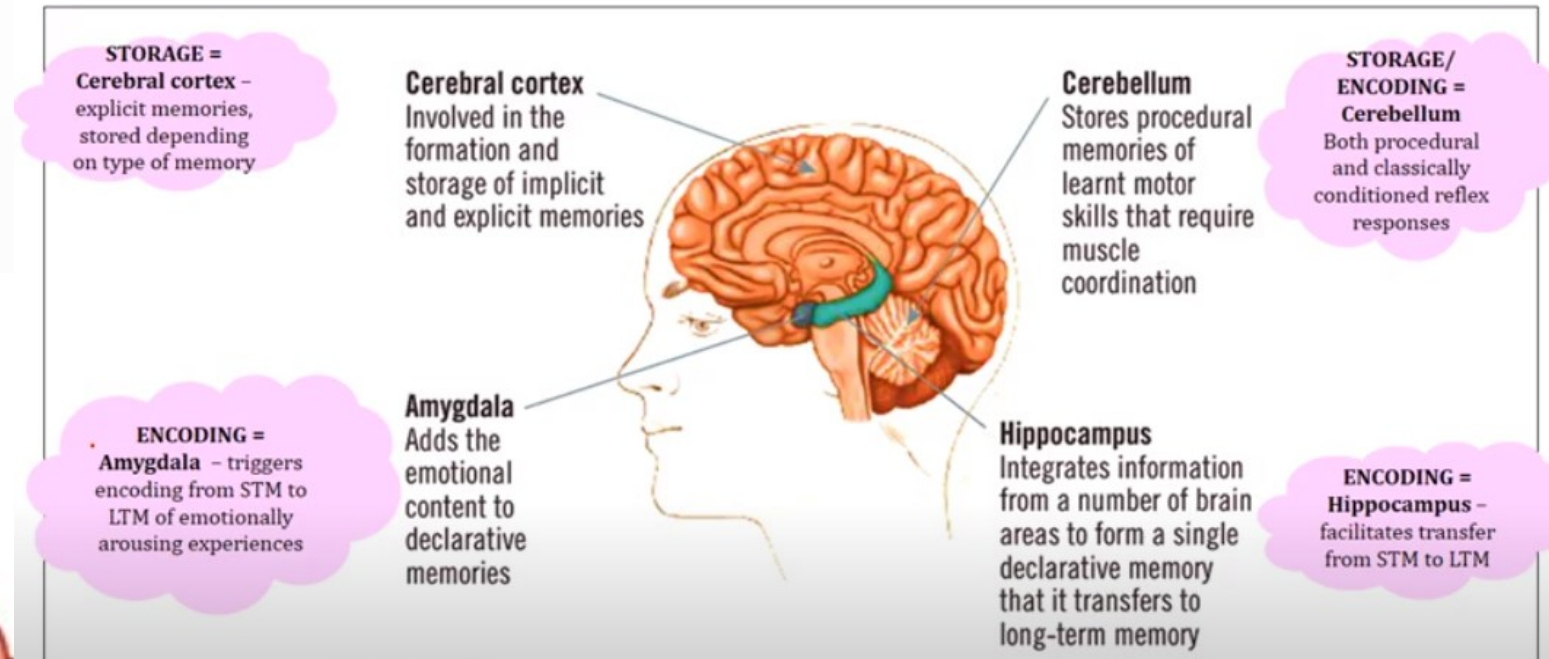
APA **KARAKTERISTIK YANG MEMBUAT PERBEDAAN** DI
SETIAP OBJEK?

CAHAYA



KENAPA KITA BISA **MENGENALI** OBJEK ?

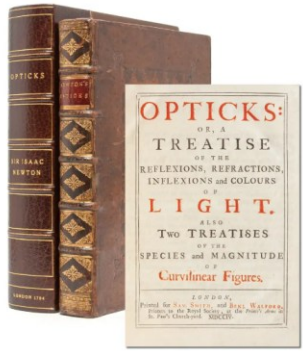
A



LILIN

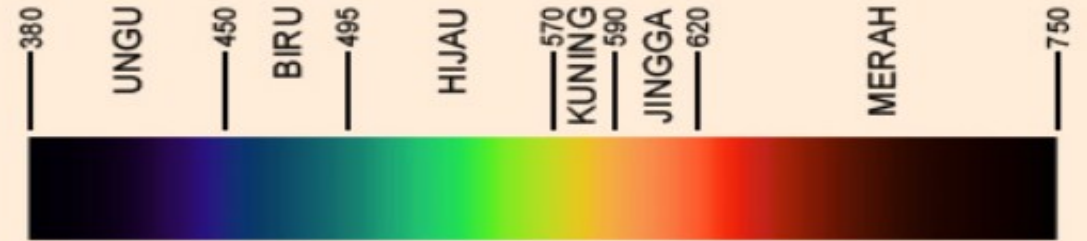
- OTAK, MEREKAM BENTUK BERBEDA SETIAP OBJEK YANG TELAH DINAMAI (BAHASA)

OPTIK



- KITA MELIHAT KARENA ADA CAHAYA
- PANJANG GELOMBANG CAHAYA YANG BERBEDA MEMBUAT KITA MELIHAT WARNA YANG BERBEDA

Warna-Warna pada Spektrum Cahaya Tampak



Warna	Panjang Gelombang	Frekuensi	Energi Foton
ungu	380–450 nm	668–789 THz	2.75–3.26 eV
biru	450–495 nm	606–668 THz	2.50–2.75 eV
hijau	495–570 nm	526–606 THz	2.17–2.50 eV
kuning	570–590 nm	508–526 THz	2.10–2.17 eV
jingga	590–620 nm	484–508 THz	2.00–2.10 eV
merah	620–750 nm	400–484 THz	1.65–2.00 eV

Pada abad ke-13 Roger Bacon mengemukakan teori bahwa pelangi dihasilkan dari proses yang sama dengan lewatnya cahaya pada kaca atau kristal.

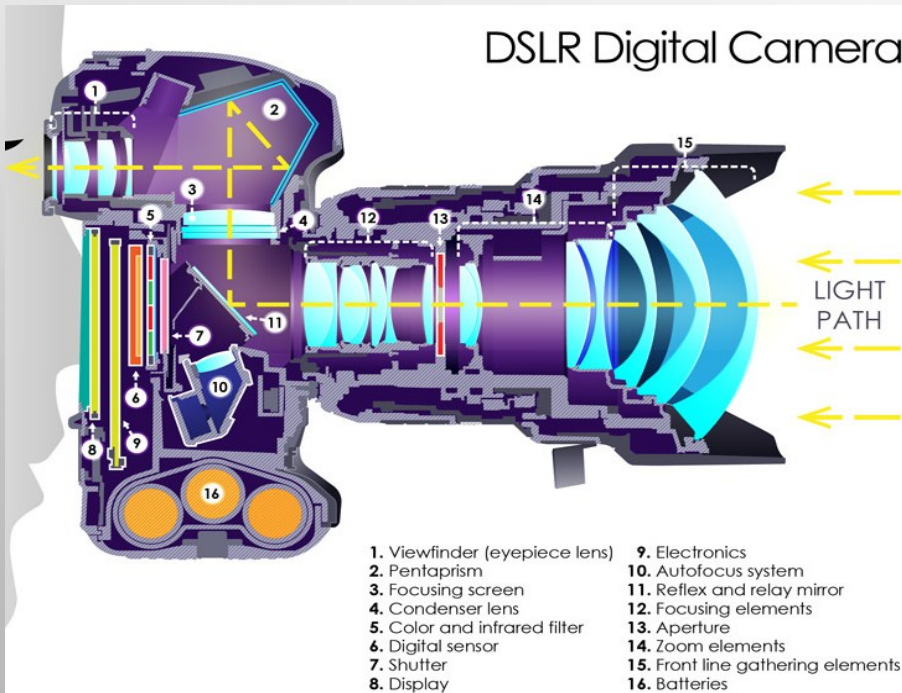
Pada abad ke-17 Isaac Newton menemukan bahwa prisma dapat menguraikan dan menyatukan cahaya putih. Newton mengemukakan fenomena tersebut dalam bukunya berjudul “Opticks”. Ia adalah orang pertama yang menggunakan kata “spektrum” yang dalam bahasa Latin berarti “penampakan” saat mendeskripsikan tentang eksperimen optiknya pada tahun 1671. Newton mengamati bahwa berkas sempit sinar matahari yang menembus prisma kaca pada suatu sudut tertentu sebagian dipantulkan dan sebagian diteruskan menghasilkan pita-pita beragam warna. Newton membagi spektrum tersebut menjadi 6 yaitu merah, jingga, kuning, hijau, biru, dan ungu.



CAMERA

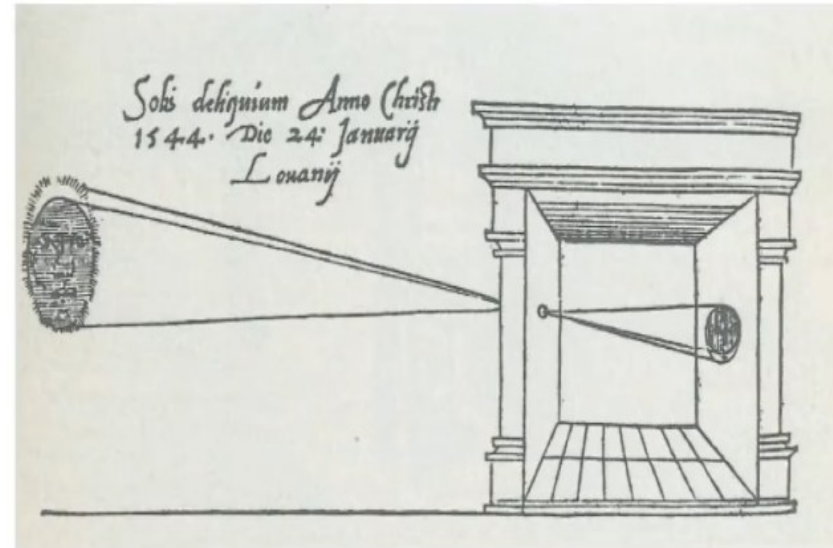
- MANUSIA MEMBUAT LENS (SENSOR) YANG MENANGKAP PANJANG GELOMBANG WARNA

DSLR Digital Camera



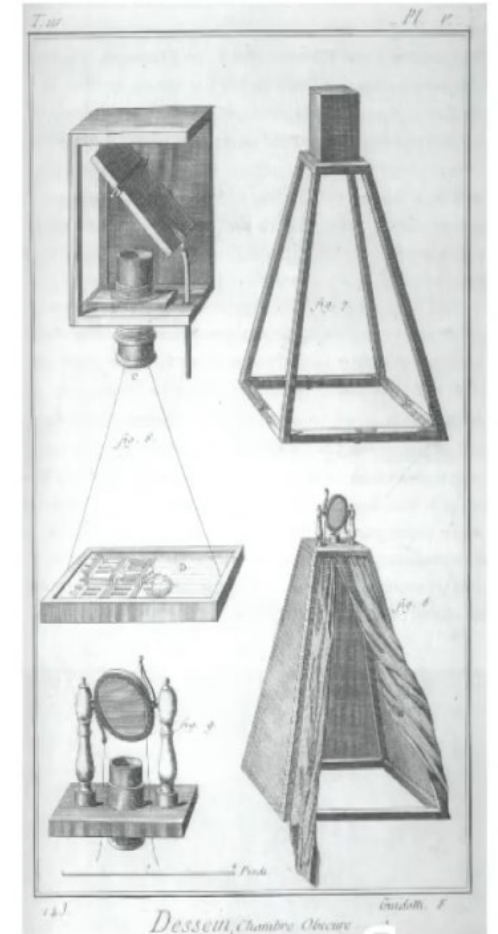
Camera Obscura

Gemma Frisius, 1545

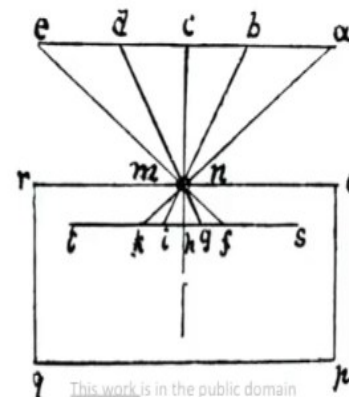


[This work is in the public domain](#)

Encyclopedie, 18th Century



[This work is in the public domain](#)

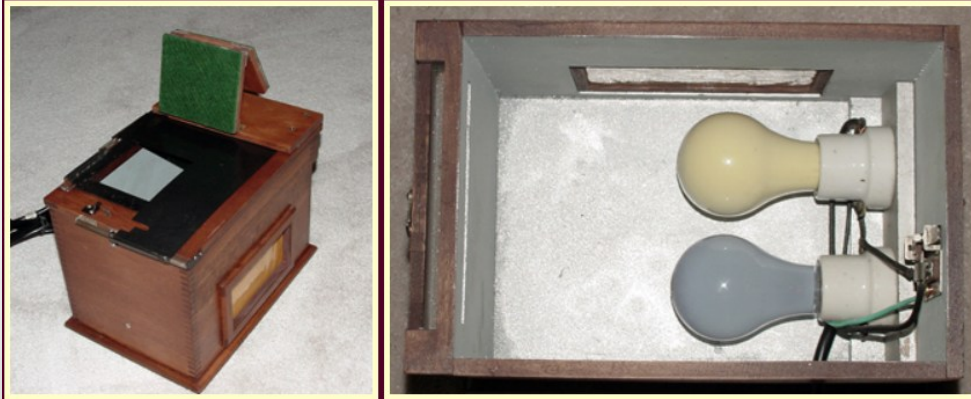


Leonardo da Vinci,
16th Century AD

[This work is in the public domain](#)

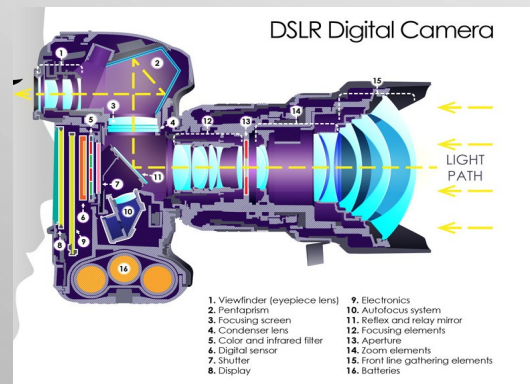
CAMERA HISTORY

KODAK 1920.

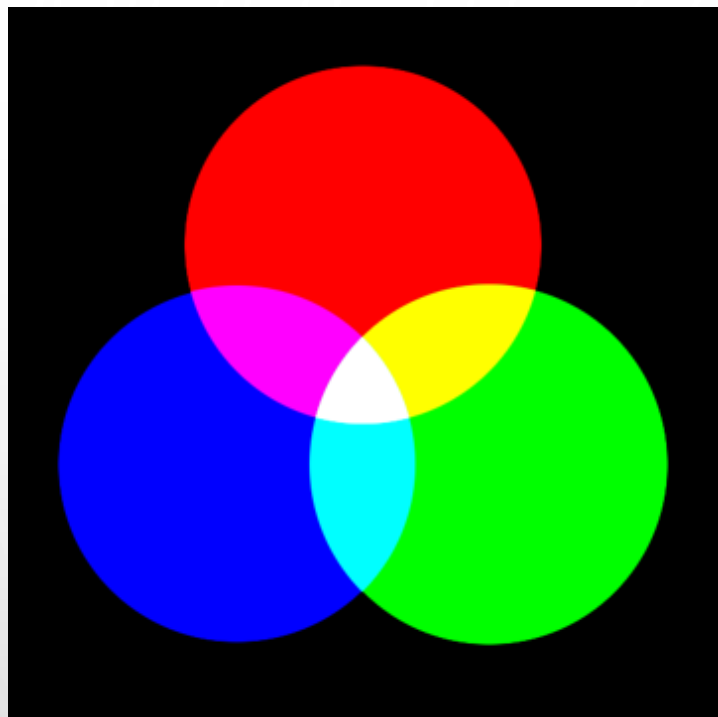


EINSTEIN / BUCKY CAMERA DESIGN - 1936. Albert Einstein and Gustav Bucky patented a camera with a photo-electric cell to operate a mechanism to reduce the intensity of the light striking the film. Two mechanisms were described: 1) a shade of varying transparency to be moved in front of the film, and 2) a mechanism to rotate a plate with various sized holes. According to Sean Johnston of the University of Glasgow, was the First to incorporate a photoelectric meter coupled to a lens aperture. [A History of Light and Color Measurement, page 229 of HTML version, Sean F. Johnston, University of Glasgow, UK](#)

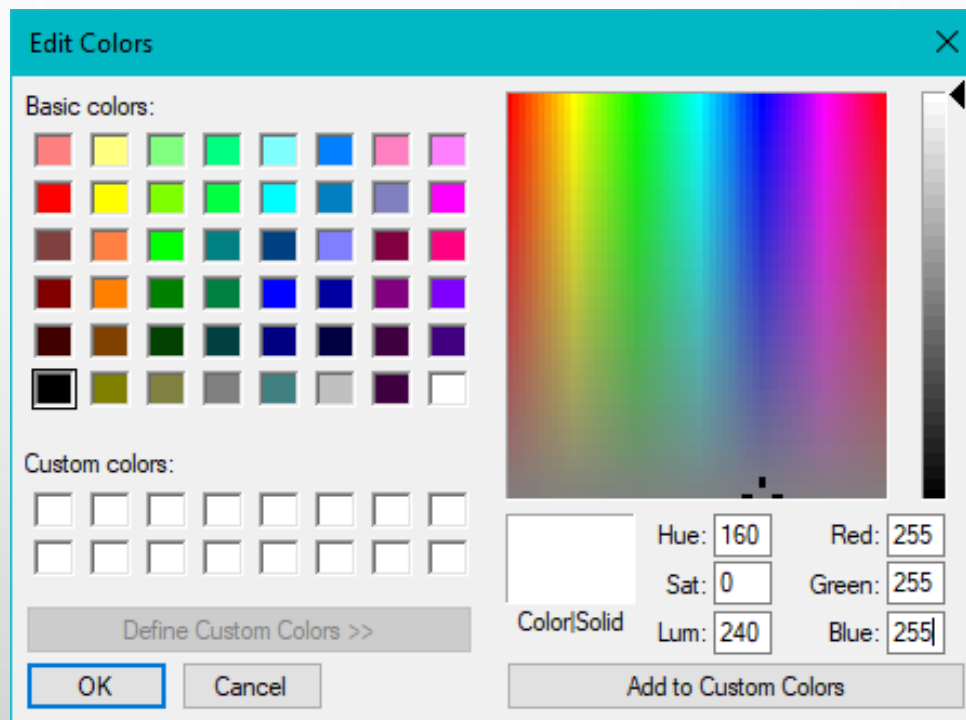
THE POLAROID
MODEL-95 - 1948



RUANG WARNA



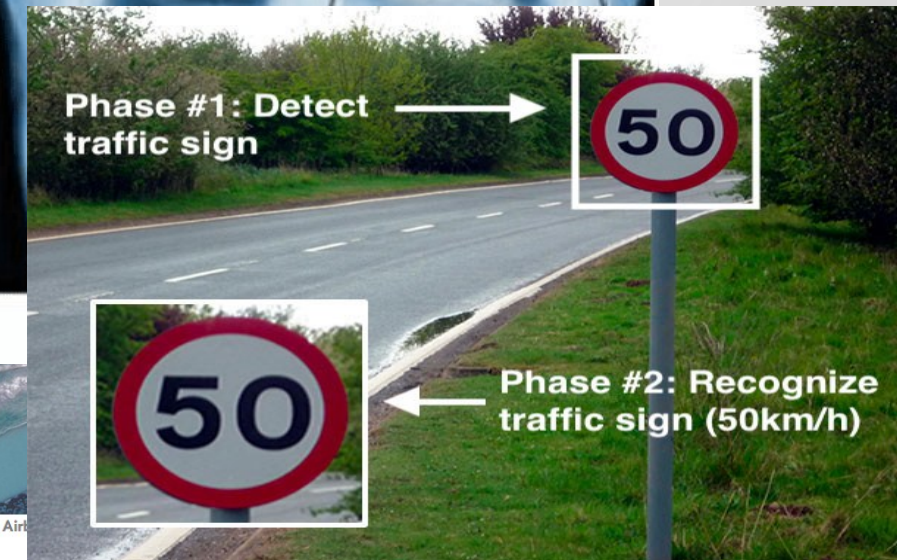
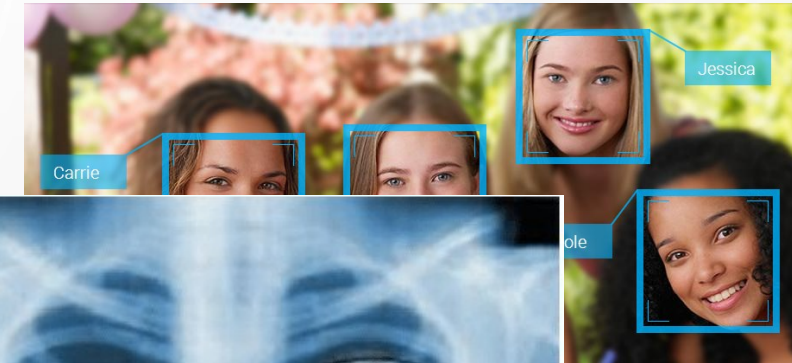
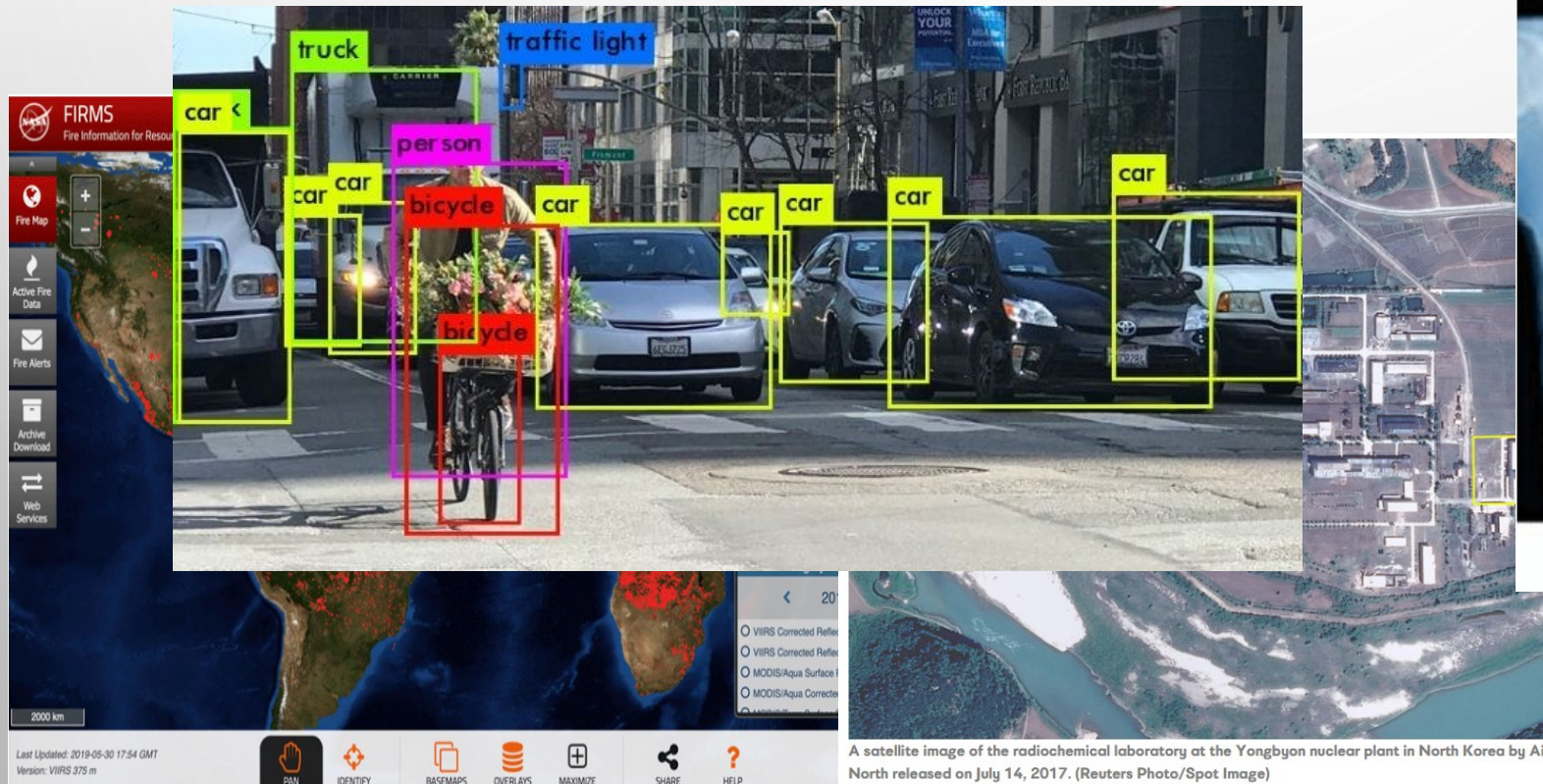
Yellow R = 255 G = 255 B = 0	Orange R = 255 G = 102 B = 0	Green R = 0 G = 255 B = 0
Cyan R = 0 G = 255 B = 255	Violet R = 204 G = 102 B = 204	White R = 255 G = 255 B = 255
Black R = 0 G = 0 B = 0	Turquoise R = 102 G = 255 B = 204	Brown R = 153 G = 102 B = 51



- WARNA-WARNA BISA DIDAPAT DARI KOMBINASI WARNA RED, GREEN DAN BLUE
- KAMERA MENGANDUNG SENSOR INTENSITAS PANJANG GELOMBANG WARNA RED, WARNA GREEN DAN WARNA BLUE

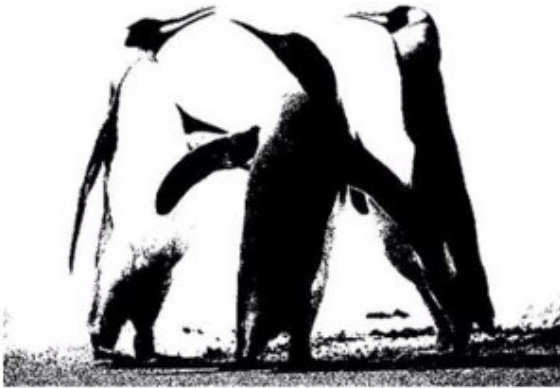
KENAPA KITA PERLU PENGOLAHAN CITRA ?

- CAMERA EVERYWHERE → BIG DATA OF IMAGE
- WE WANT COMPUTER LIKE HUMAN



DIGITAL IMAGE TYPE

Binary



Grayscale



Color



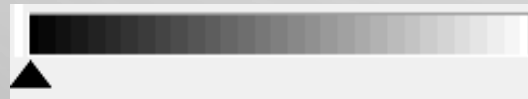
	Hue: 0	Red: 255
	Sat: 240	Green: 0
Color Solid	Lum: 120	Blue: 0

	Hue: 80	Red: 0
	Sat: 240	Green: 255
Color Solid	Lum: 120	Blue: 0

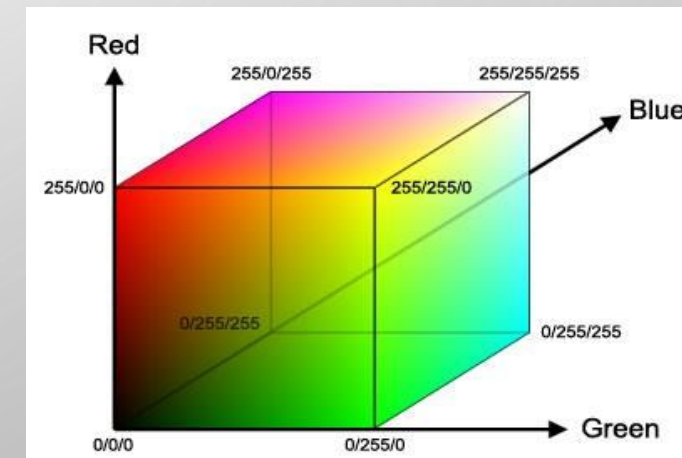
	Hue: 160	Red: 0
	Sat: 240	Green: 0
Color Solid	Lum: 120	Blue: 255

- PIKSEL 0 DAN 1
(HITAM/PUTIH)

- PIKSEL 0-255
- 1 CHANNEL
 - GRAY, 0 - 255



- PIKSEL 0-255
- 3 CHANNEL :
 - RED, 0 - 255
 - GREEN, 0 - 255
 - BLUE, 0 - 255

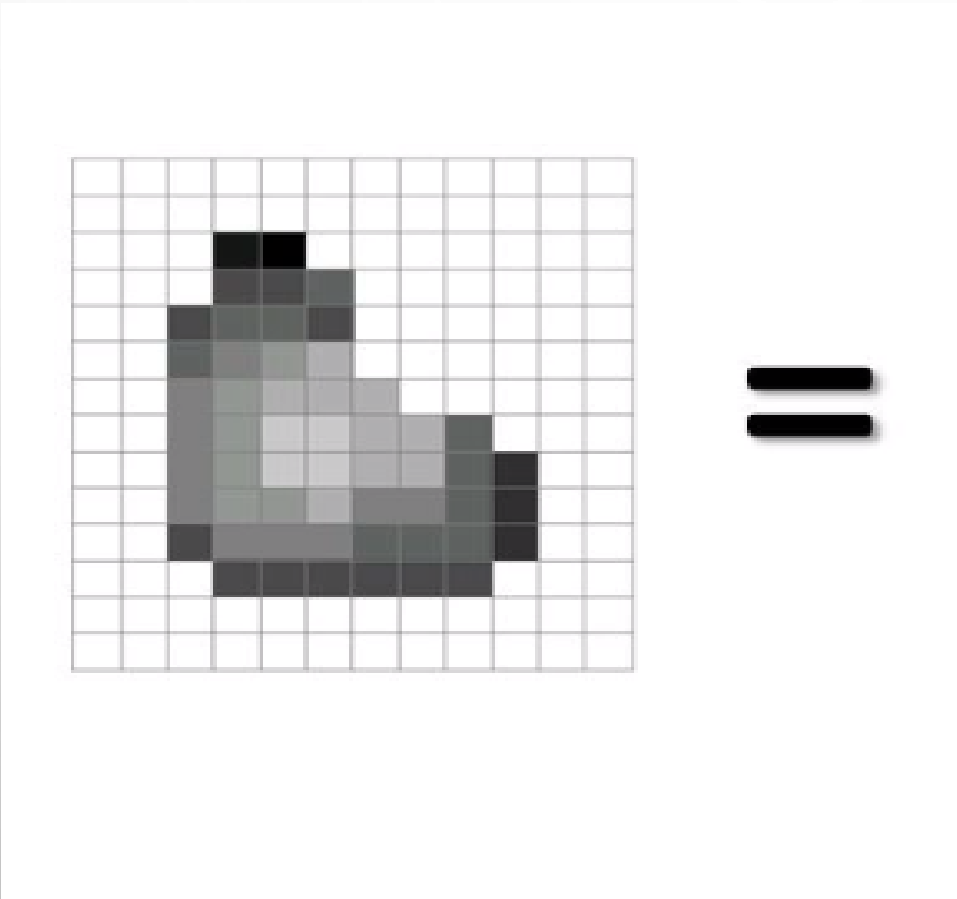


- 
- A diagram showing a dark blue, irregular shape representing a pixel. The shape has a jagged top edge with several peaks and valleys. The entire shape is outlined in a thin blue line. The text "Pixel (Picture Element)" is written in white, sans-serif font, centered within the dark blue area.

Pixel (Picture Element)

GRAY IMAGE

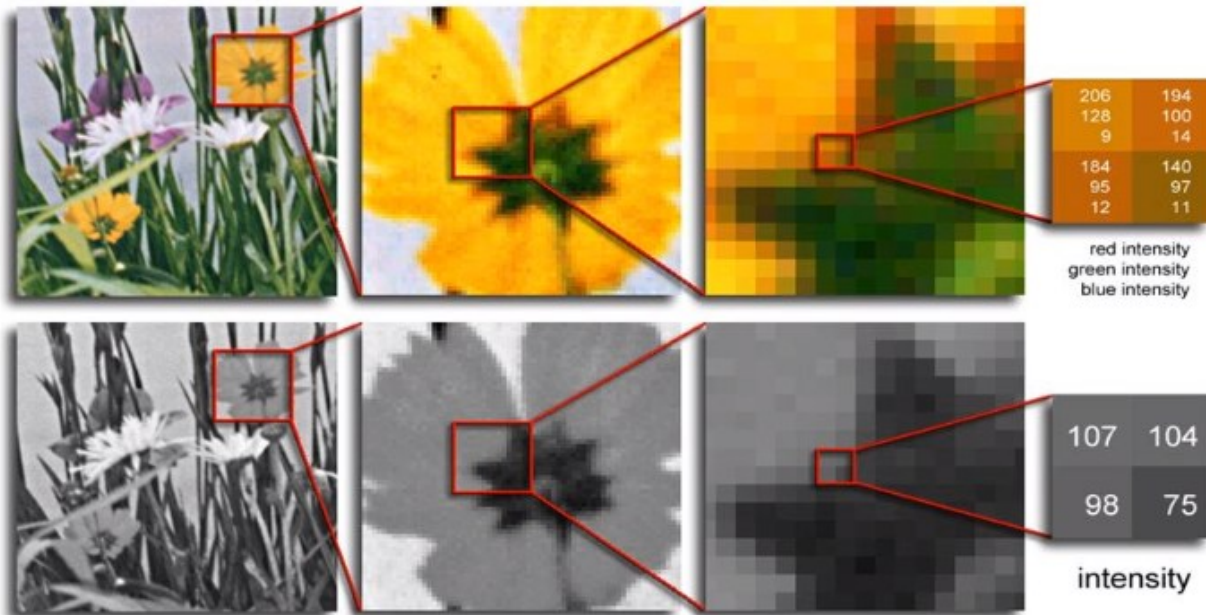
- $255 = \text{PUTIH}$
- $0 = \text{HITAM}$

[illegible][illegible]

COLOR IMAGE

Color

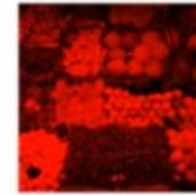
3 Channels – Red, Green, Blue



RGB



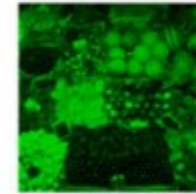
R



Red



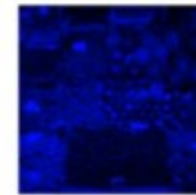
G



Green



B



Blue



107 104
98 75

intensity

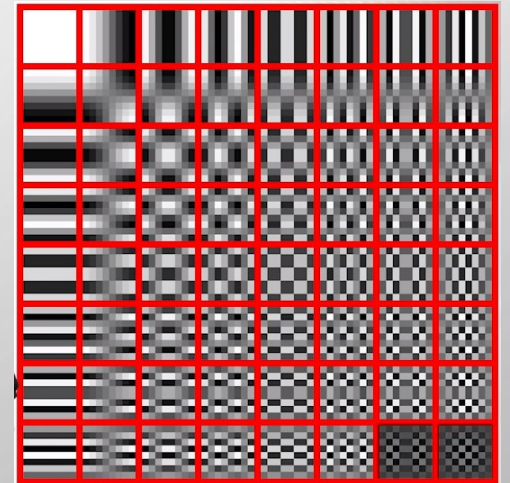
FORMAT FILE CITRA DIGITAL

- RAW IMAGE
- BMP (DIB), BITMAP IMAGE (DEVICE INDEPENDENT BITMAP)
- **JPEG, JOINT PHOTOGRAPHIC EXPERT GROUP**
- GIF, GRAPHIC INTERCHANGE FORMAT
- **PNG, PORTABLE NETWORK GRAPHICS**
- TIFF, TAGGED IMAGE FILE FORMAT
- SVG, VECTOR GRAPHICS

JPEG – LOSSY COMPRESSION



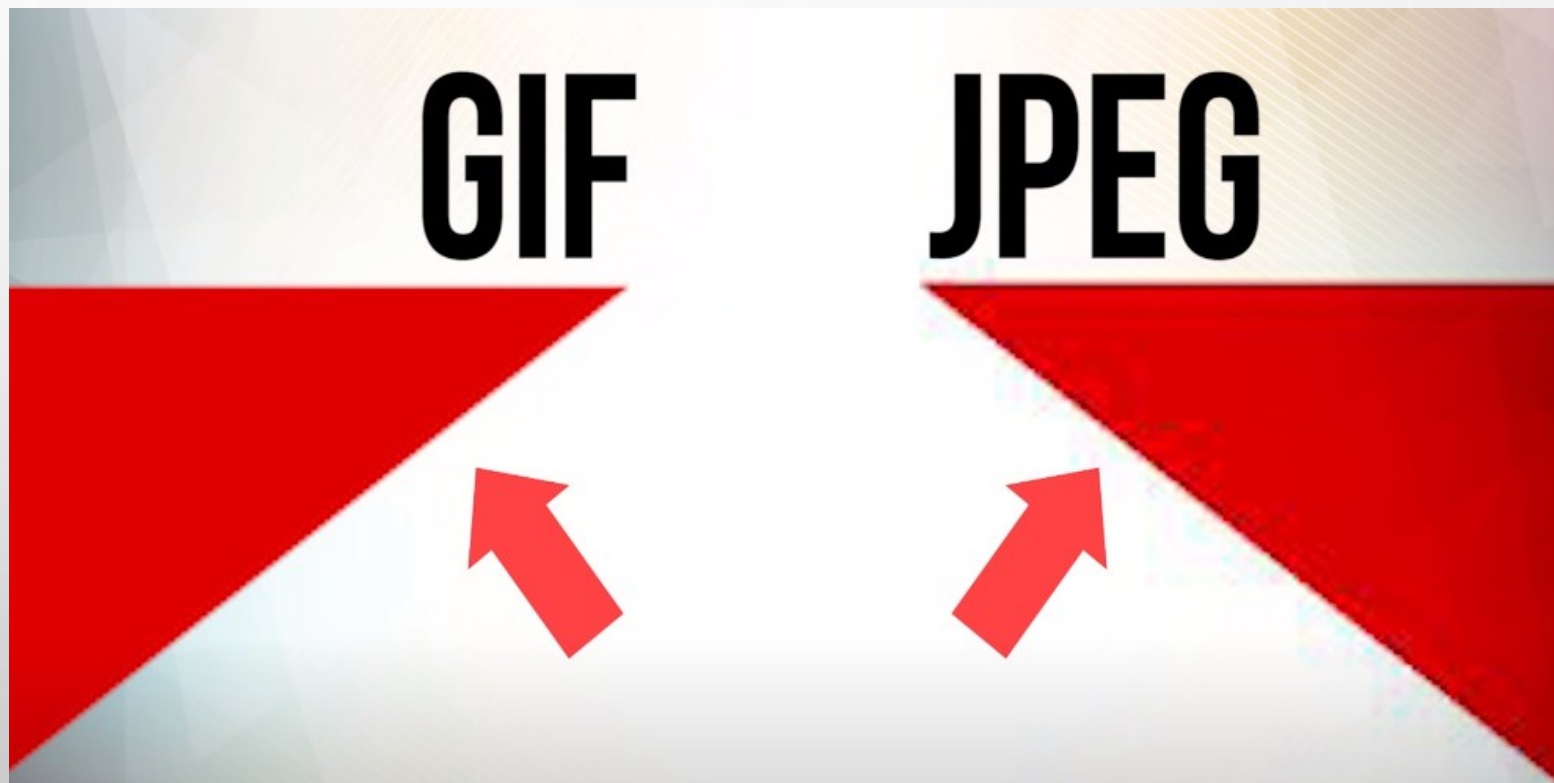
- REDUCE FILE SIZE SAMPAI 90%.
- 8X8 BLOCK PIXEL DICARI FREQUENCY TERTINGGI DIGANTI DENGAN 16 BLOCK PATTERN YG SUDAH DISIAPKAN.





GIF VS JPEG

- SUPPORT GAMBAR ANIMASI





- GIF HANYA MENYIMPAN 256 WARNA

GIF



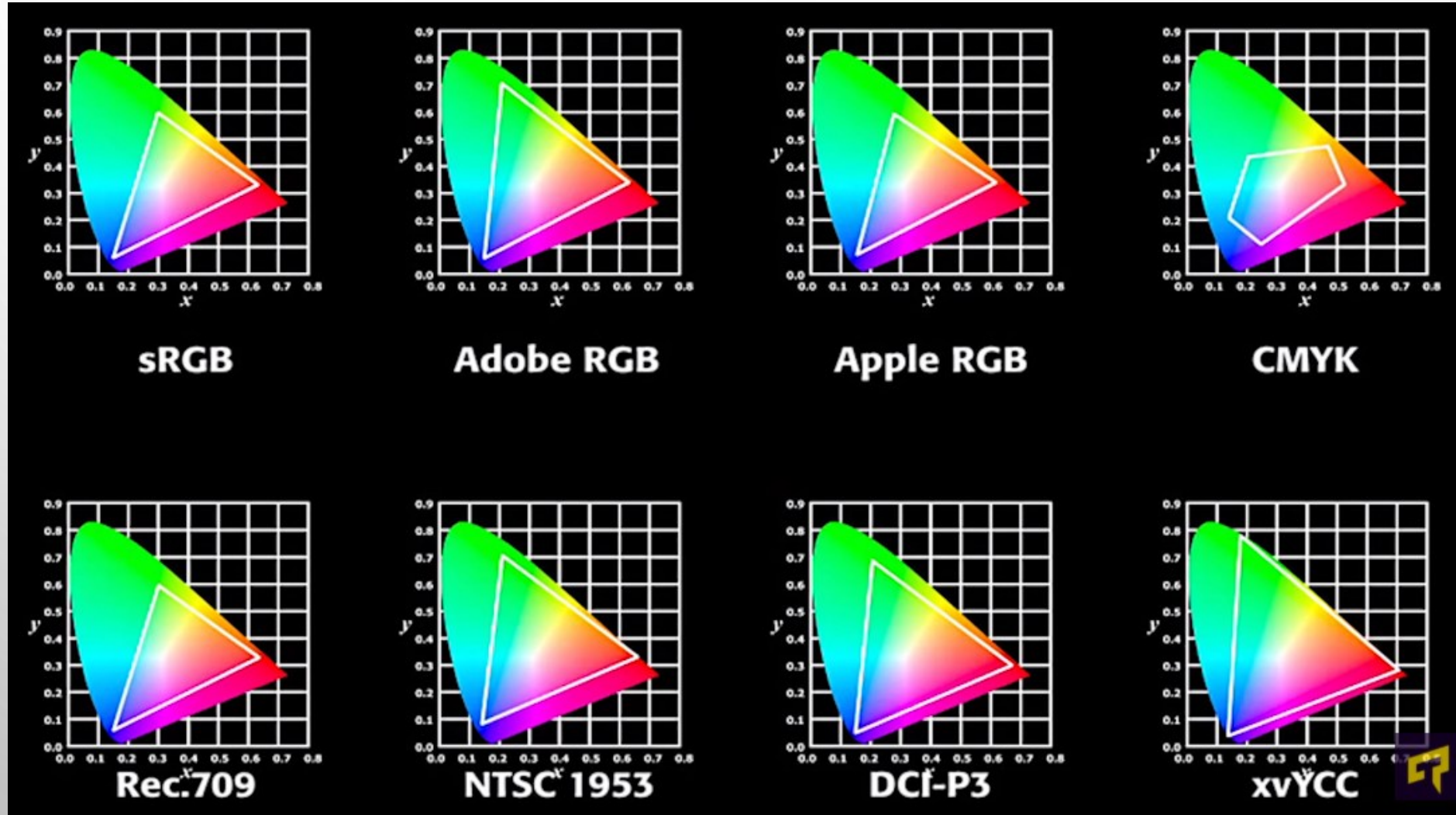
PNG – LOSS LESS COMPRESSION

- SUPPORT TRANSPARANCY
- GOOD FOR TEXT AND SHAPES

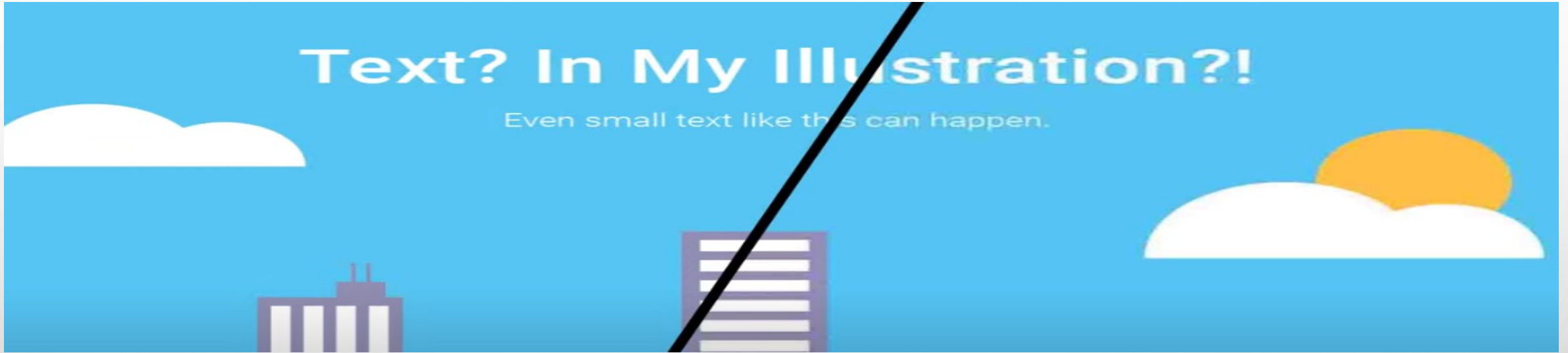


TIFF – PROFESIONAL PHOTOGRAFER FOR PRINTING

- HUGE FILE SIZE,
SUPPORT
COMPRESSED AND
UNCOMPRESSED
IMAGE



JPG VS PNG VS GIF



For website designer :

Use JPEG for Scene Photos

Use PNG for Image with Text or Shapes

Use GIF for Logo



SAMPLING AND QUANTIZATION

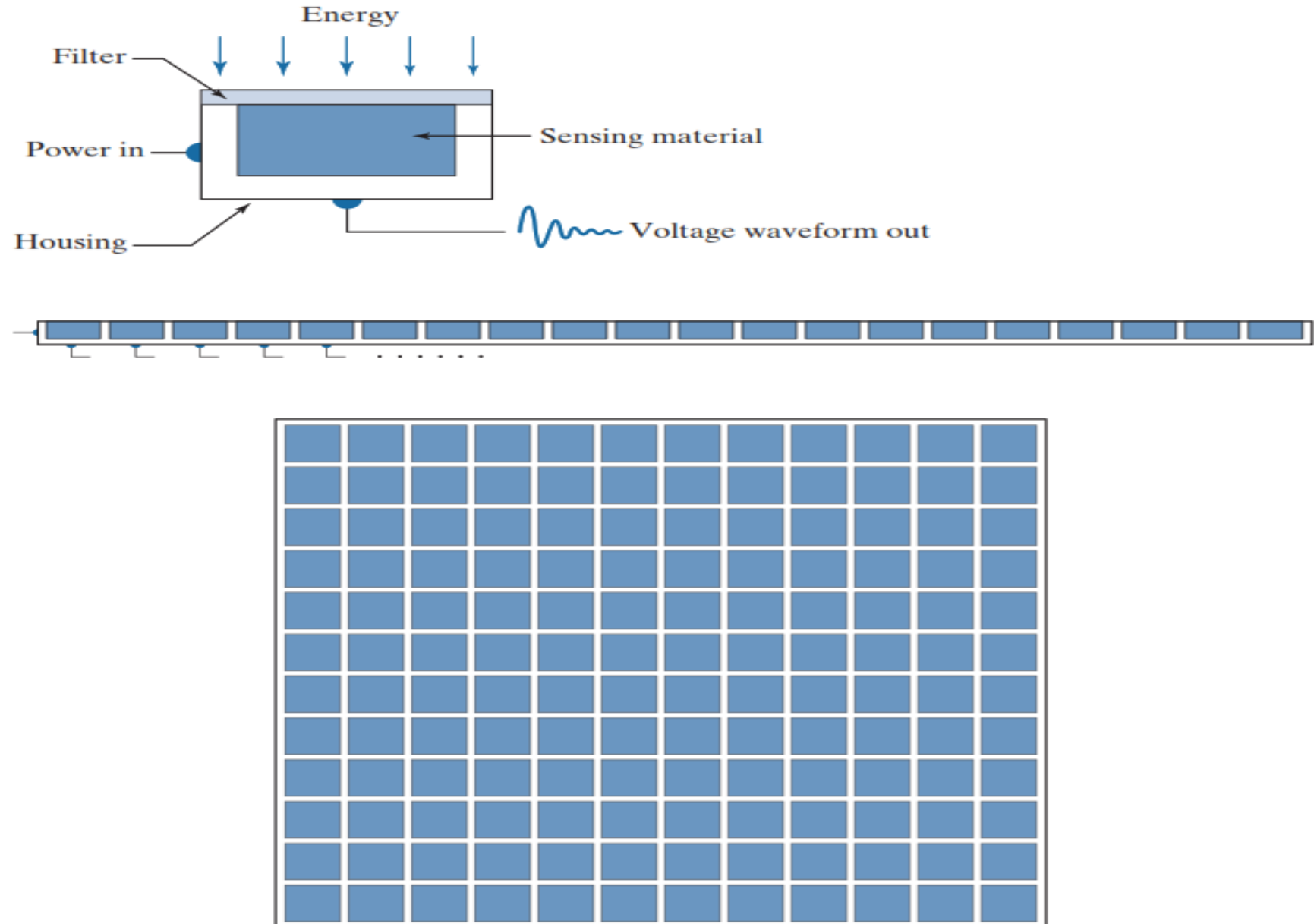
a
b
c

FIGURE 2.12

(a) Single sensing element.

(b) Line sensor.

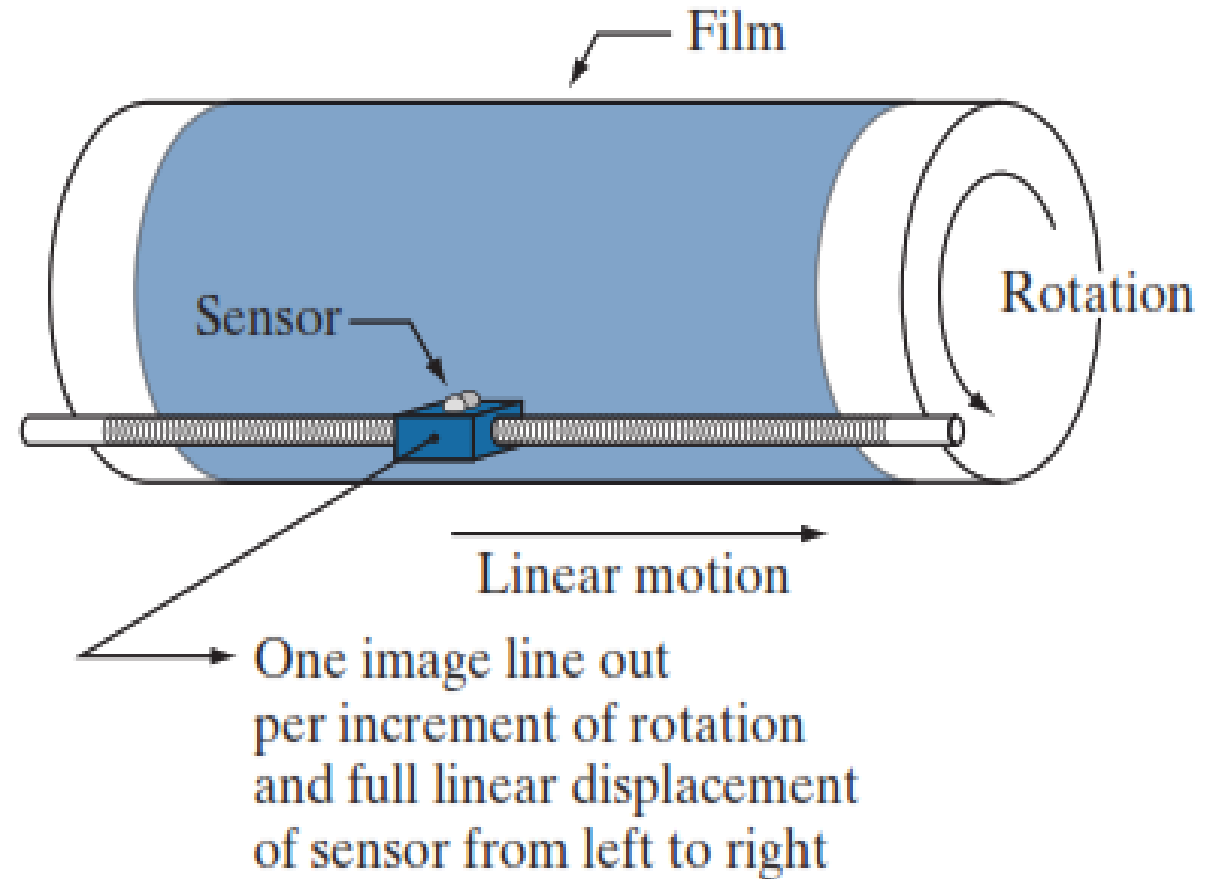
(c) Array sensor.



SAMPLING AND QUANTIZATION

FIGURE 2.13

Combining a single sensing element with mechanical motion to generate a 2-D image.

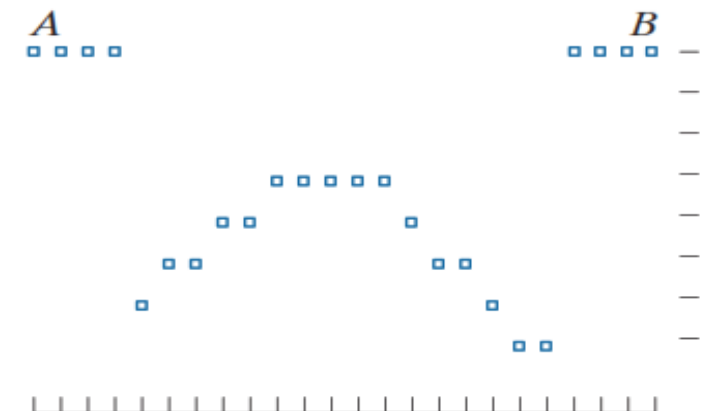
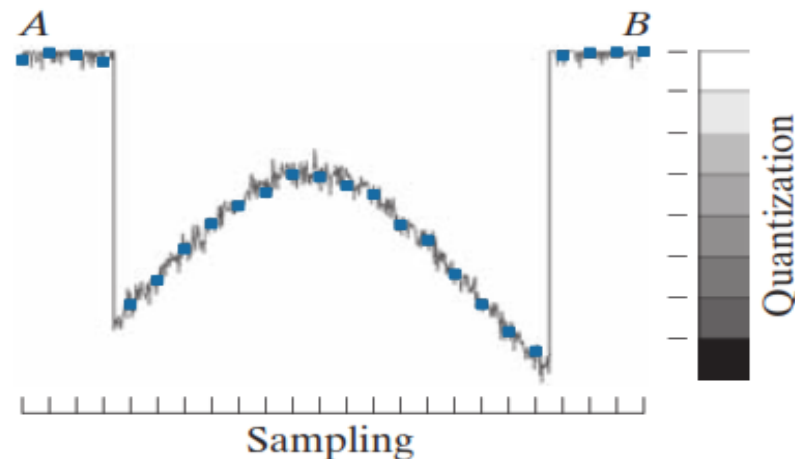
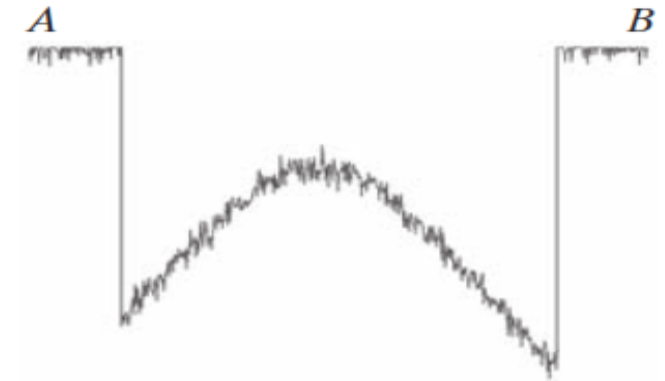
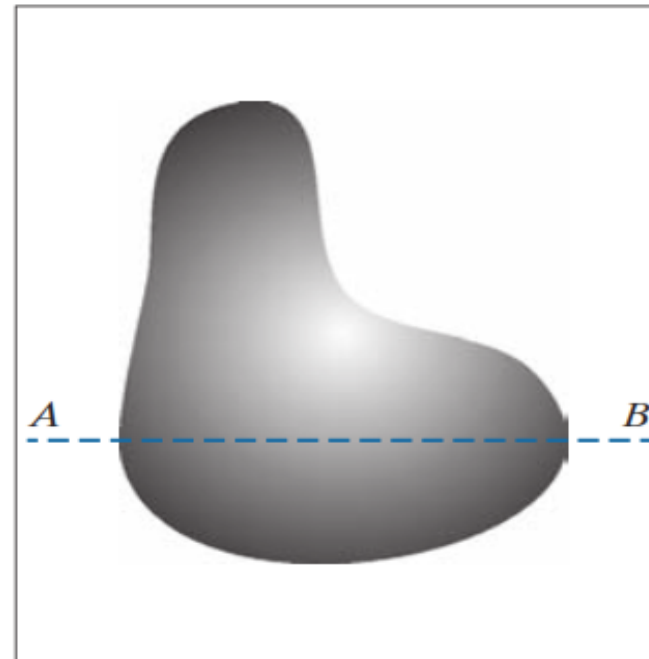


SAMPLING AND QUANTIZATION

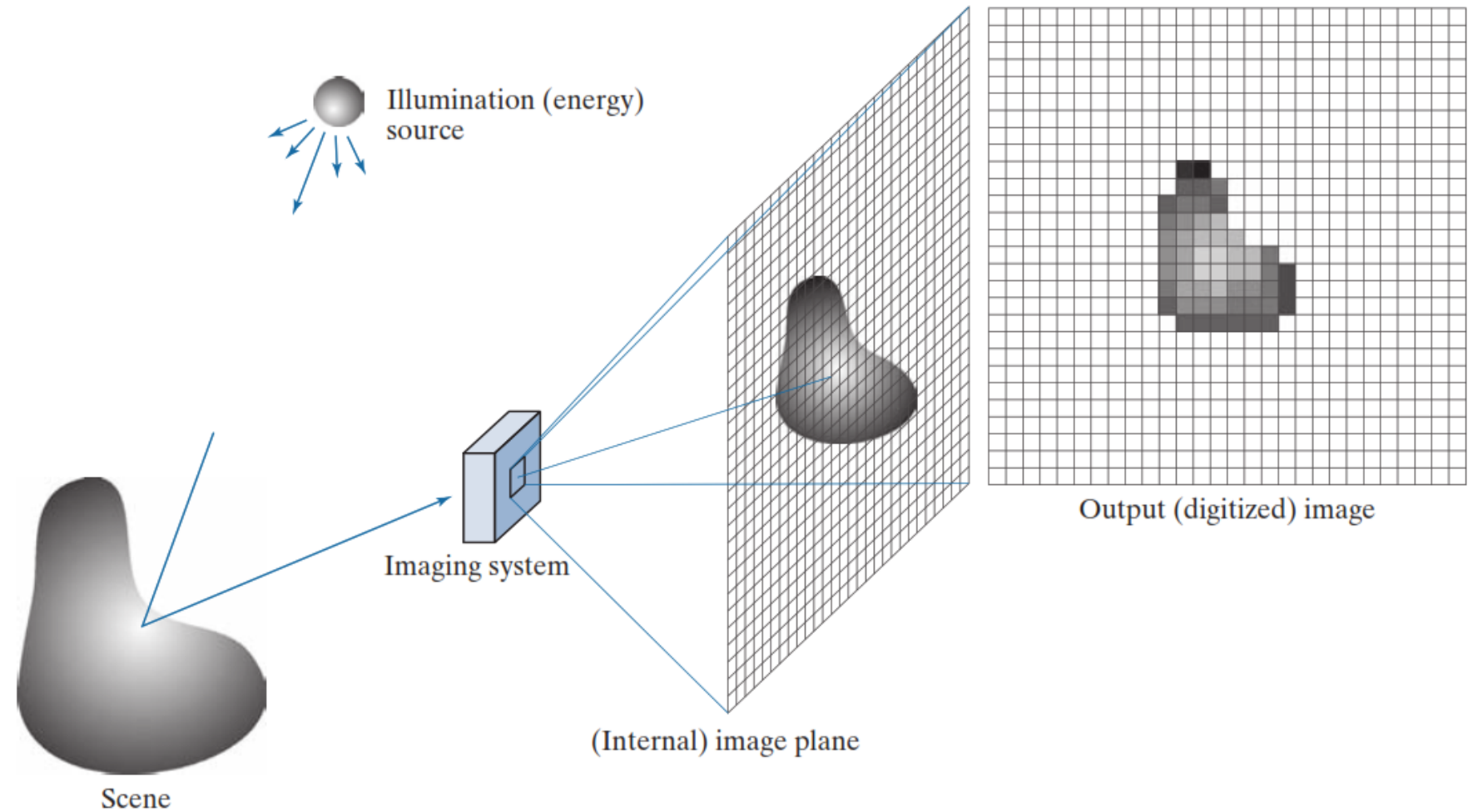
a	b
c	d

FIGURE 2.16

(a) Continuous image. (b) A scan line showing intensity variations along line AB in the continuous image. (c) Sampling and quantization. (d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).



SAMPLING AND QUANTIZATION



a
b c d e

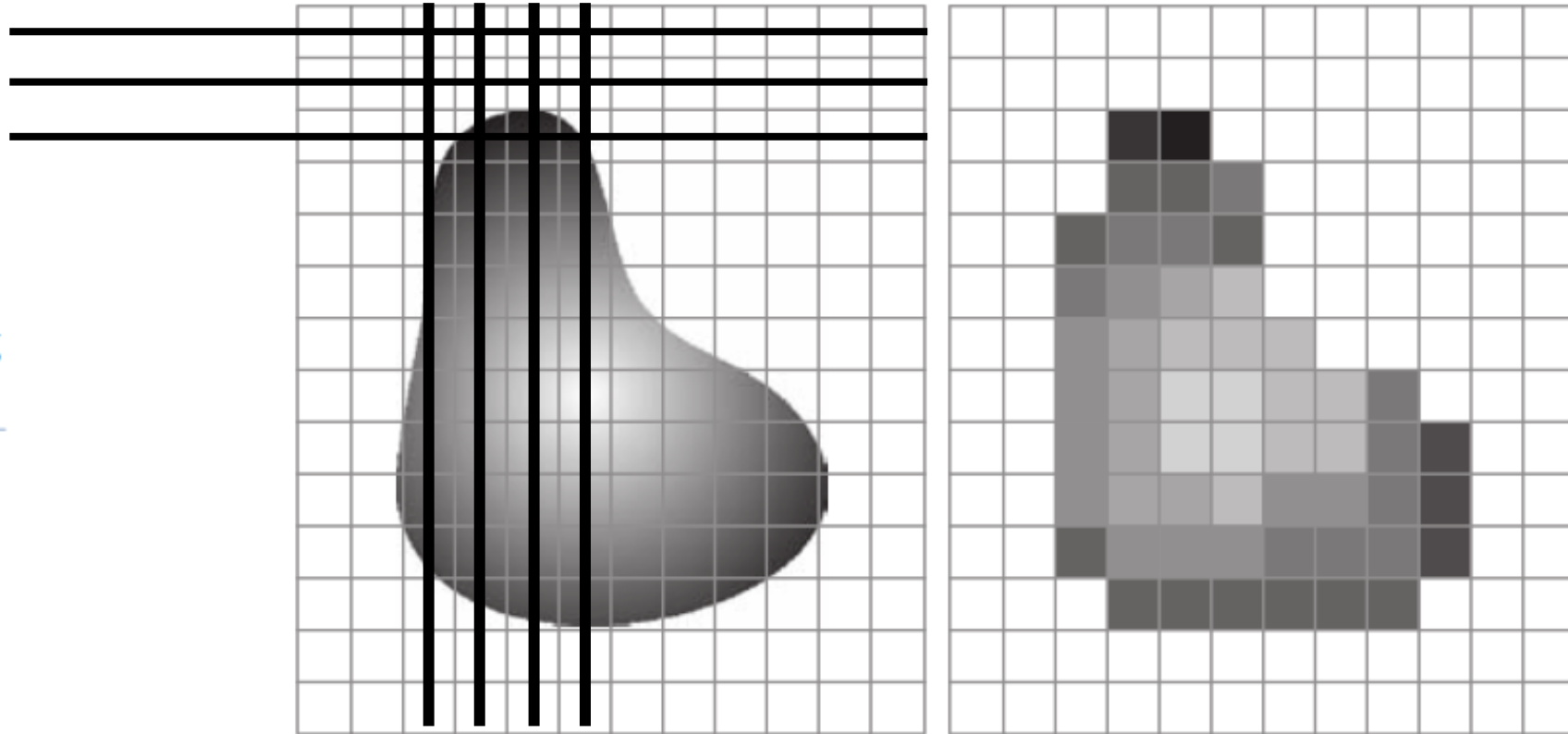
FIGURE 2.15 An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

SAMPLING AND QUANTIZATION

a b

FIGURE 2.17

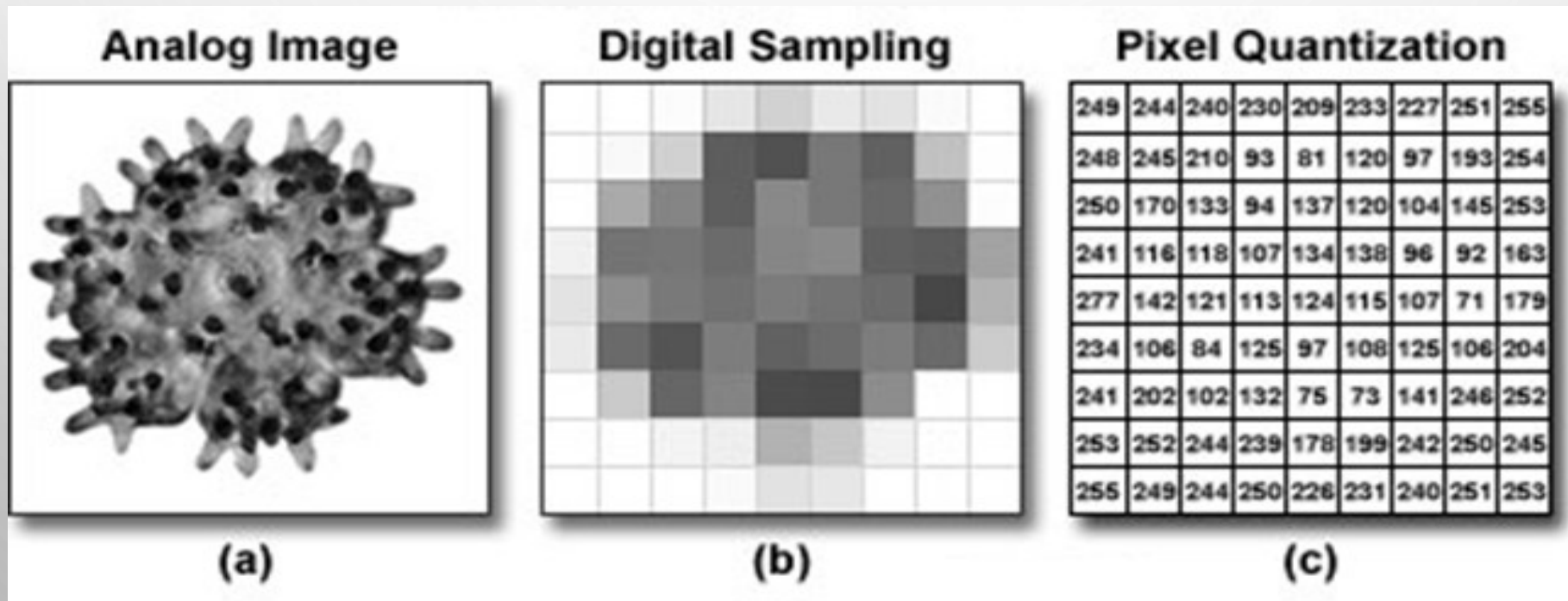
(a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.



$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

SAMPLING AND QUANTIZATION

- SAMPLING → MAPPING KOORDINAT CITRA (X,Y), CANVAS/MATRIKS. JUMLAH PIXELS.
- QUANTIZATION → MAPPING VALUE, RANGE RGB/GRAY.



MEGAPIXEL (MP)

Number of Megapixels	Image Dimensions
2	1740 x 1160
3	2160 x 1440
4	2450 x 1633
6	3000 x 2000
8	3504 x 2336
10	3872 x 2592
12	4256 x 2832
14	4608 x 3072
16	4928 x 3264
18	5184 x 3456
24	6016 x 4000
32	6920 x 4613
36	7360 x 4912
50	8688 x 5792
61	9504 x 6336